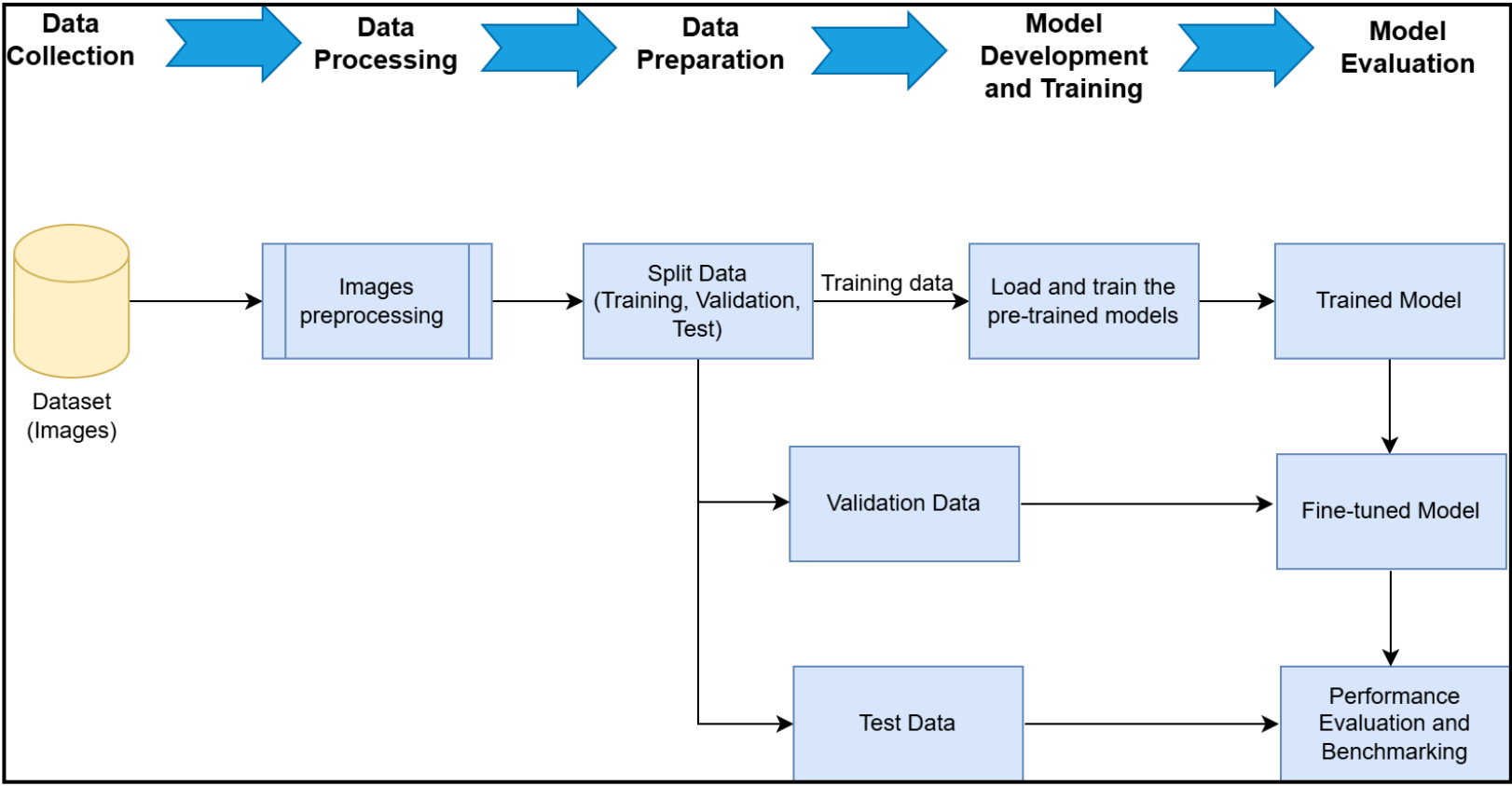


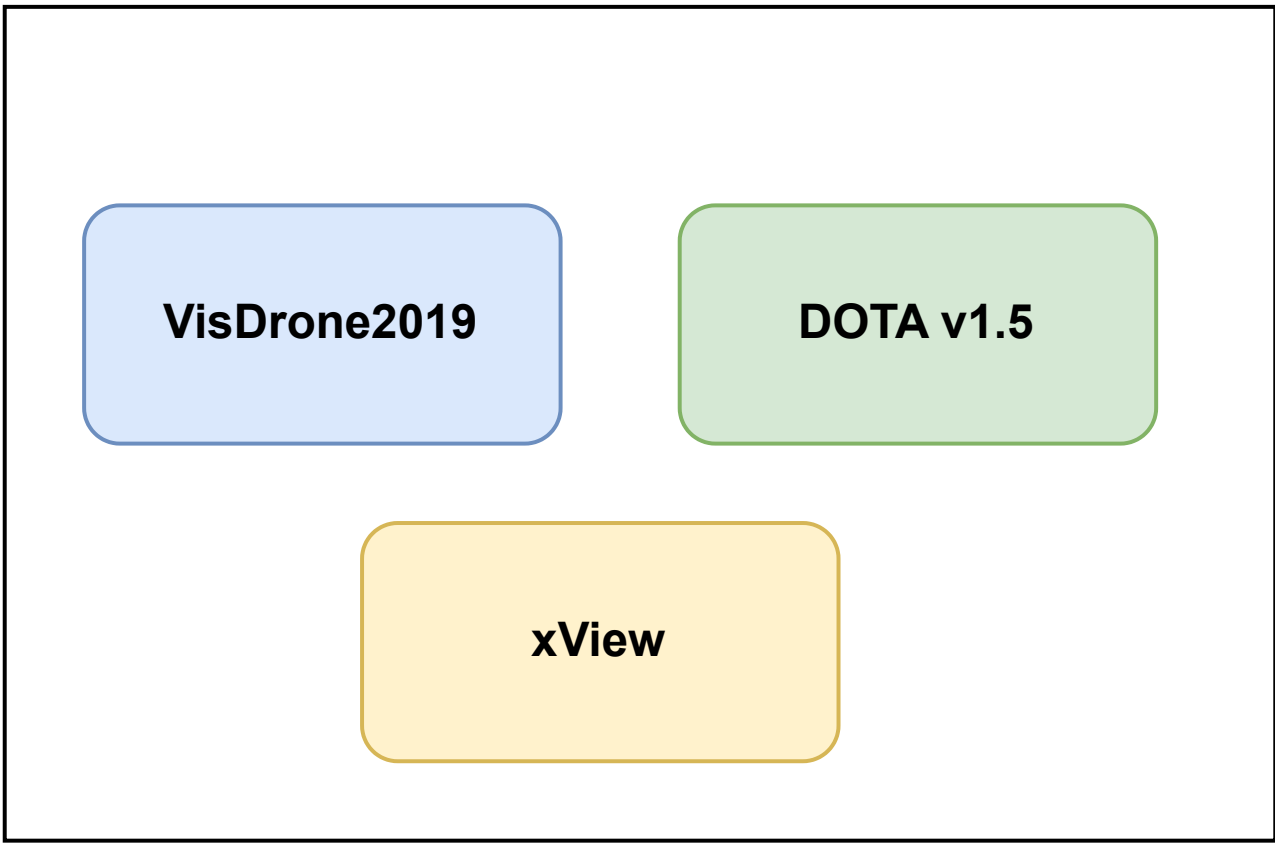
Small-Object Detection at the Edge: A Pareto-Efficient Benchmark of Lightweight YOLO Models on UAV and Overhead Datasets

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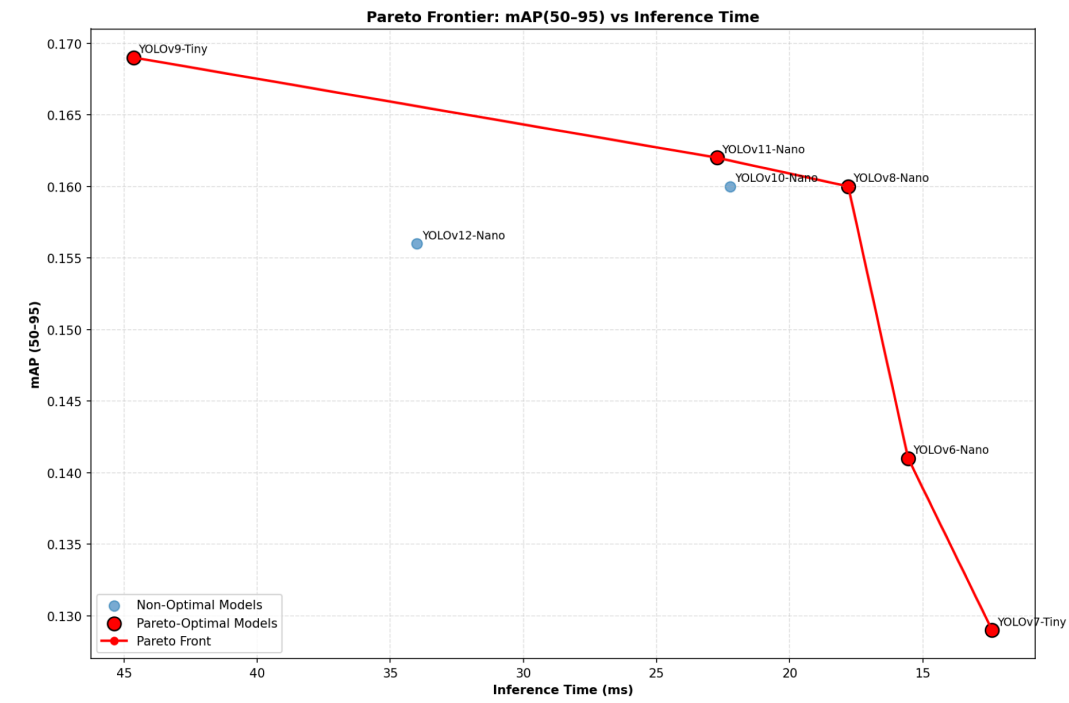
1 Beacom College of Computer & Cyber Sciences, Dakota State University, Madison, SD 57042 USA
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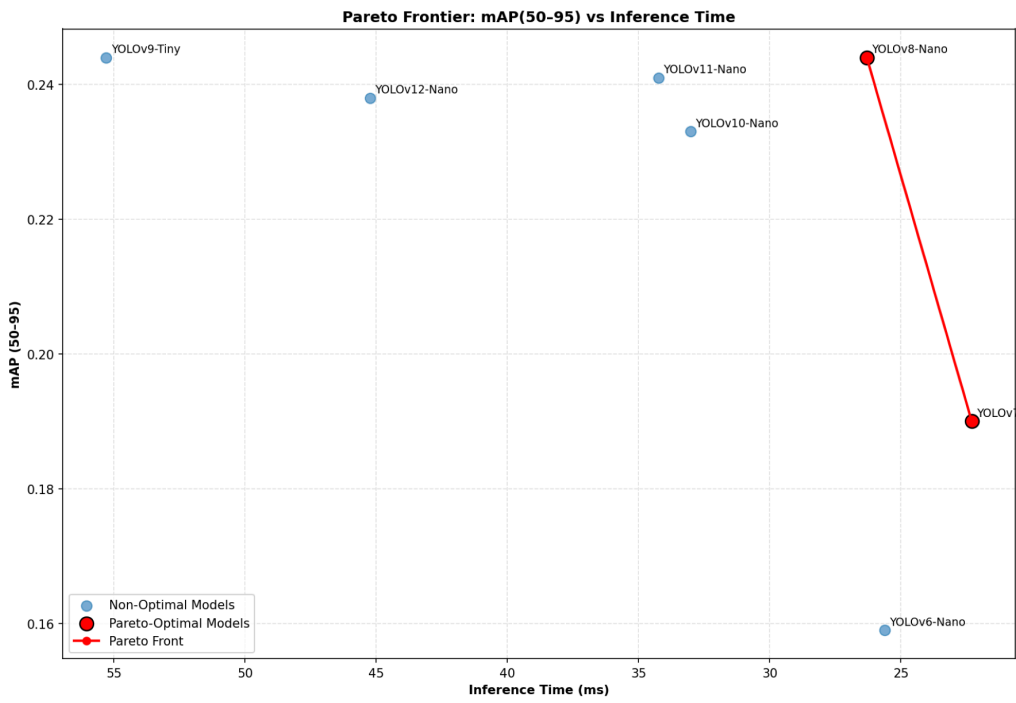
Experimental Flow Diagram



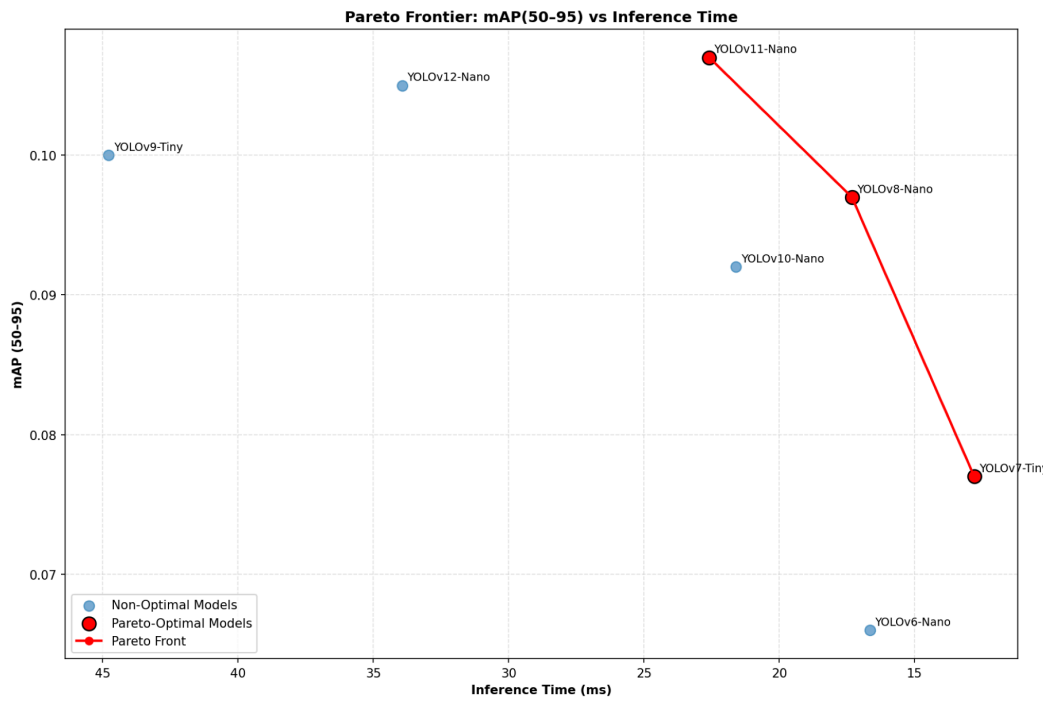
Benchmarked Datasets



(a) VisDrone2019



(b) DOTA v1.5



(c) xView

Pareto Frontier Evaluation

Key Findings:

- YOLOv8-Nano and YOLOv9-Tiny: Best overall accuracy.
- YOLOv6-Nano and YOLOv10-Nano: Optimal trade offs between speed and memory.
- TensorRT format delivers 3x-5x speedup vs PyTorch format.
- Pareto Front: No single lightweight model dominates speed, memory, and accuracy.